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SYNOPSIS Presentation

on

The Therapeutic Potential and Risks of Tablet-Based Apps for Pain and Anxiety Management during Pediatric Venipuncture

by

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Project Overview

What is the Problem?

Pediatric venipuncture (blood test) causes:

- Pain
- Anxiety
- Emotional distress
- Fear of future medical procedures

Proposed Solution

Using tablet-based applications (games, videos, relaxation apps) as a non-pharmacological intervention.

Why is it Important?

- Improves child comfort
 - Reduces medical trauma
 - Supports SDG 3: Good Health & Well-being
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Project Objectives

The main objectives are:

1. To evaluate the effectiveness of tablet-based apps in reducing pain.
2. To assess anxiety reduction during pediatric venipuncture.
3. To identify potential risks and limitations.
4. To recommend safe integration in clinical settings.

Success will be measured through:

- Pain scores (VAS/FLACC)
 - Anxiety levels
 - Behavioral observations
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Project Scope

Included:

- Study of digital health apps
- Analysis of clinical research evidence
- Pain & anxiety measurement models
- Risk and ethical evaluation

Excluded:

- Drug-based pain management
 - Adult patient studies
 - Surgical procedures
 - Key Modules:
 - Literature Review Module
 - Data Analysis Module
 - Risk Assessment Module
 - Conclusion & Recommendation Module
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Methodology

Research Approach:

- Literature-based analytical study
- Comparative analysis of control vs experimental groups

Tools & Technologies:

- Digital Health (mHealth)
 - Statistical Analysis Tools
 - Research Databases
 - Data Visualization Tools (Excel/SPSS)
 - Development Model:
 - Waterfall Model (Sequential Research Process)
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Hardware & Software Requirements

Hardware:

- Laptop/PC
- Tablet Device (for app demonstration)
- Internet Access

Software:

- MS PowerPoint
 - MS Word
 - Research databases (Google Scholar, PubMed)
 - Statistical tools (Excel/SPSS)
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Timeline

Phase	Duration
• Topic Selection	Week 1
• Literature Review	Week 2–4
• Data Collection	Week 5–6
• Analysis	Week 7
• Report Writing	Week 8
• PPT Preparation	Week 9

Milestones:

- Abstract Approval
 - Mid Review
 - Final Submission
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Project Outcome

Final Output:

Research Paper

Expected Contributions:

- Evidence-based evaluation
- Clinical recommendations
- Improved pediatric care strategies

Long-term Impact:

- Better hospital experience for children
 - Integration of digital health tools in clinical practice
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